

<< SkyTeam >> Use Cases

Use Case 1: Starting the Game

Players start the game on a computer by selecting "Start Game".

Actors:

- Player 1, Player 2

Pre-conditions:

- This is a new game from round 1.

Functional Requirements:

- System prompts players to select roles.

Use Case 2: Choosing roles

Description: Players choose the roles of Co-pilot or Pilot.

Actors:

- Player 1 and Player 2

Pre-conditions:

- Game is played by 2 players.

Functional Requirements:

- Start round 1

Post-conditions:

- Each player knows their roles and rules of the game.

Use Case 3: Rolling the Dice

Players roll the dice for current round.

Actors:

- Pilot and Co-pilot

Pre-conditions:

- This is a new round.

Functional Requirements:

- System generates 4 random dice values for each player.

Post-conditions:

- Player knows the dice values and strategy.
- System displays dice values in the dice screen.

Use Case 4: Dice Placement

Players place dice on designated spots on the control panel by moving the dice from roll area to the board game.

Actors:

- Pilot and Co-pilot

Pre-conditions:

- It is the players turn to play the game.

Functional Requirements:

- Player drags a dice and places it on the board.
- System checks if it is a valid place for that dice value.
- System updates and saves the move of the player.

Post-conditions:

- System moves the player token according to the game rules.

Use Case 5: Axis control

Players place the dice on the airplane axis dice holder to adjust the airplane's tilt and maintain balance. The sum of the values entered by both players will be considered as a final value to adjust the axis.

Actors:

- Pilot and Co-pilot

Pre-conditions:

- Both players have placed 1 dice each in the axis placeholder.

Functional Requirements:

- System calculates difference between both values entered by the players.
- System increases the axis value on the side of player with higher dice value.

Alternate Flow:

- If the axis value is more than 2 on any players side, then game ends.
- System displays "Game Over" message with the reason of failure.

Post-conditions:

- System moves the axis token with respect to saved axis value.

Use Case 6: Engine Speed

Players adjust the airplane's speed by placing dice in the engine placeholder. The total value of dice in the engine area determines the airplane's forward movement speed.

Actors:

- Pilot and Co-pilot

Pre-conditions:

- Both players have placed 1 dice each in the engine placeholder.

Functional Requirements:

- System adds the dice values of both players.
- System compares total value with aerodynamics marker on the speed guage.
- If the value is greater than orange speed marker, plane advances by 2 tracks value.
- If the value is between blue and orange speed marker, Plane advances by 1 track value.
- If the value is less than blue speed marker, plane does not move.

Alternate Flow:

- If there are planes on the track when advancing, then game ends.
- System displays "Game Over" message with the reason of failure.

Post-conditions:

- System moves the track by 1 or 2 space.

Use Case 7: Landing gear

Player deploys the landing gear to increase the aerodynamics drag to slow down the plane speed before approaching the Airport. Each time a dice is placed on one of the 3 placeholder for the first time, the blue speed marker moves forward by 1.

Actors:

- Pilot (Player 1)

Pre-conditions:

- Only Pilot can deploy the landing gears.
- The gear is not deployed before.
- Gears can be deployed in any order.

Functional Requirements:

- Pilot places the dice on the landing gears placeholders labeled as 1/2, 3/4, 5/6.
- System checks if the pilot has placed the correct dice in the placeholder.
- If incorrect, system will put the dice back to its original place.
- If correct, system will increase the blue speed marker value by 1 unit.

Alternate Flow:

- If the gear is already deployed and has a green light, dice goes back to original place.

Post-conditions:

- System will move the blue marker 1 unit ahead.
- System will turn on green light for that landing gear placeholder.

Use Case 8: Flaps

Player deploys the flaps to increase the plane's lift and wind resistance to slow down the plane speed before approaching the Airport. When a dice is placed on the flaps placeholder in a specific order for the first time, the orange speed marker moves forward by 1.

Actors:

- Co-pilot (Player 2)

Pre-conditions:

- Only Co-pilot can deploy the flaps.
- The flap is not deployed before.
- Flaps can only be deployed in top to bottom order.
- Previous Flap is already deployed if it is not the first flap.

Functional Requirements:

- Co-pilot places the dice on the orange placeholders labeled 1/2, 3/4, 5/6. in a specific order.
- System checks if the pilot has placed the correct dice in the placeholder.
- If incorrect, system will revert back to player input stage.
- If correct, system will increase the orange speed marker value by 1 unit.

Alternate Flow:

- If the flap is already deployed and has a green light, dice goes back to original place.

- **Post-conditions:**

- System will move the orange marker 1 unit ahead.
- System will turn on green light for that flap placeholder.

Use Case 9: ATC Radio

Players place dice in the radio placeholder to clear airplane traffic ahead on the approach track.

Actors:

- Pilot and Co-pilot

Pre-conditions:

- The Radio placeholder for player using it should be empty.
- Pilot can only use it once per round.
- Co-pilot can use it twice per round.

Functional Requirements:

- Players places a dice on Radio placeholder.
- System checks for airplanes at the corresponding distance(dice value) from the airplane's current position
- System removes 1 airplane from that distance.
- System updates the number of airplanes on the approach track.

Alternate Flow:

- If there is no airplane on that distance, do nothing.

Post-conditions:

- System moves 1 airplane token from the distance value.

Use Case 10: Brakes

The pilot uses brakes to help reduce speed when landing in the final round.

Actors:

- Pilot

Pre-conditions:

- Only Pilot can apply brakes.
- The brake level is not applied before.
- Brakes can only be applied increasing order.
- Previous brake level is already applied if it is not the first level.

Functional Requirements:

- Pilot places the dice on the blue placeholders labeled 2, 4, 6 in a specific order.
- System checks if the pilot has placed the correct dice in the placeholder.
- If incorrect, system will revert back to player input stage.
- If correct, system will increase the red brake marker value by 1 unit.

Alternate Flow:

- If the brake lever is already pulled and has a green light, do nothing.

Post-conditions:

- System will move the red marker 1 unit ahead.
- System will turn on green light for that brake placeholder.

Use Case 11: Re-roll Tokens

Players decide to use re-roll tokens to re-roll one or more dice left in their dice screens.

Actors:

- Pilot and Co-pilot

Pre-conditions:

- There should be a re-roll token available.
- There should be unused dices with at least 1 player.

Functional Requirements:

- Players choose to use re-roll token.
- System generates new random dice values for the selected dices by the players.
- System saves new dice values that were generated.
- Reduce the re-roll token value by 1 if any available.

Post-conditions:

- Player knows the new dice values.
- System switches new the dice values in the dice screen.
- Remove the re-roll token from the placeholder.
- Update the available re-roll token value.

Use Case 12: Buy Coffee Tokens

A coffee token can be used to change the dice value by 1 unit. The tokens should be purchased by dice before using it in the future.

Actors:

- Pilot and Co-pilot

Pre-conditions:

- Tokens should be available in the inventory for purchase.
- Maximum of 3 tokens can be purchased on the board.

Functional Requirements:

- Players places a dice in the concentration placeholder to get a coffee token.
- If the number of coffee tokens are less than 3, system increases the number of available token by 1.

Alternate flow:

- If there are already 3 tokens on the board, system does nothing.

Post-conditions:

- System adds a coffee token in the coffee placeholder.

Use case 13: Using Coffee Tokens

A coffee token can be used to change the dice value by 1 unit. The tokens should be purchased by dice before using it in the future. A coffee token can be used by either player if available on the board.

Actors:

- Pilot and Co-pilot

Pre-conditions:

- Tokens should be available in the coffee placeholder.
- Player has unassigned dice available in the dice screen.

Functional requirements:

- Players choose to use a coffee token.
- System prompts to increase or decrease the selected dice value
- Player chooses to increase or decrease the dice value.
- System changes the dice value.
- System updates the number of available coffee tokens.

Post-conditions:

- System displays the new dice value on the dice screen.
- System removes a coffee token from the placeholder.

Use Case 14: Check next round conditions

After players complete their turns, the system checks for all the conditions required to move to the next round. If the conditions are not met, the game ends.

Actors:

- System

Pre-conditions:

- All the dice have been assigned by both players.

Functional Requirements:

- System checks for Engine and Axis placeholders.
- If either of Engine or Axis placeholders are empty, game ends with "Game Over" message.
- The system lowers the plane altitude by 1 unit space at the end of the round.
- System resets some functionalities to carry forward with the next round.

Alternate Flow:

- If plane has not reached to the final airport track yet before the final round, the game ends (Crash landing)
- If the plane has reached the destination before the final round, the game ends (Overshot the location)

Post-conditions:

- System moves the altitude track lower by 1 unit place.
- System displays the board functionalities for next round.
- If it was the final round, System shows game win message.